Homework #1

Section 7.1

Let

$$\mathbf{A} = \begin{bmatrix} 0 & 2 & 4 \\ 6 & 5 & 5 \\ 1 & 0 & -3 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 0 & 5 & 2 \\ 5 & 3 & 4 \\ -2 & 4 & -2 \end{bmatrix}$$

$$\mathbf{C} = \begin{bmatrix} 5 & 2 \\ -2 & 4 \\ 1 & 0 \end{bmatrix}, \quad \mathbf{D} = \begin{bmatrix} -4 & 1 \\ 5 & 0 \\ 2 & -1 \end{bmatrix},$$

$$\mathbf{E} = \begin{bmatrix} 0 & 2 \\ 3 & 4 \\ 3 & -1 \end{bmatrix}$$

$$\mathbf{u} = \begin{bmatrix} 1.5 \\ 0 \\ -3.0 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} -1 \\ 3 \\ 2 \end{bmatrix}, \quad \mathbf{w} = \begin{bmatrix} -5 \\ -30 \\ 10 \end{bmatrix}.$$

Find the following expressions, indicating which of the rules in (3) or (4) they illustrate, or give reasons why they are not defined.

9. $3\mathbf{A}$, $0.5\mathbf{B}$, $3\mathbf{A} + 0.5\mathbf{B}$, $3\mathbf{A} + 0.5\mathbf{B} + \mathbf{C}$

$$3A = 3\begin{bmatrix} 0 & 2 & 4 \\ 6 & 5 & 5 \\ 1 & 0 & -3 \end{bmatrix} = \begin{bmatrix} 0 & 6 & 12 \\ 18 & 15 & 15 \\ 3 & 0 & -9 \end{bmatrix}$$

$$0.5B = 0.5\begin{bmatrix} 0 & 5 & 2 \\ 5 & 3 & 4 \\ -2 & 4 & -2 \end{bmatrix} = \begin{bmatrix} 0 & 2.5 & 1 \\ 2.5 & 1.5 & 2 \\ -1 & 2 & -1 \end{bmatrix}$$

$$3A + 0.5B = \begin{bmatrix} 0 & 8.5 & 13 \\ 20.5 & 16.5 & 17 \\ 2 & 2 & -10 \end{bmatrix}$$

$3A + 0.5B + C$

No possible: $3A + 0.5B$ is 3×3
C is 3×2
Incompatible sizes!

12. $(\mathbf{C} + \mathbf{D}) + \mathbf{E}$, $(\mathbf{D} + \mathbf{E}) + \mathbf{C}$, $0(\mathbf{C} - \mathbf{E}) + 4\mathbf{D}$, $\mathbf{A} - 0\mathbf{C}$

$$(C+D)+E = \left(\begin{bmatrix} 5 & 2 \\ -2 & 4 \\ 1 & 0 \end{bmatrix} + \begin{bmatrix} -4 & 1 \\ 5 & 0 \\ 2 & -1 \end{bmatrix}\right) + \begin{bmatrix} 0 & 2 \\ 3 & 4 \\ 3 & -1 \end{bmatrix} = \begin{bmatrix} 1 & 5 \\ 6 & 8 \\ 6 & -2 \end{bmatrix}$$

$(C+D)+E = (D+E)+C$ Rule 3B

$0(C-E)+4D = 4D$ Rule 3C

$$4D = 4\begin{bmatrix} -4 & 1 \\ 5 & 0 \\ 2 & -1 \end{bmatrix} = \begin{bmatrix} -16 & 4 \\ 20 & 0 \\ 8 & -4 \end{bmatrix}$$

$A-0D$ No Possible: A is 3x3, 0D is 3x2. Incompatible sizes

13. $(2 \cdot 7)\mathbf{C}$, $2(7\mathbf{C})$, $-\mathbf{D}+0\mathbf{E}$, $\mathbf{E}-\mathbf{D}+\mathbf{C}+\mathbf{u}$

$(2 \cdot 7)C = 2(7C) = 14C$ Rule 4C

$$14C = 14\begin{bmatrix} 5 & 2 \\ -2 & 4 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 70 & 28 \\ -28 & 56 \\ 14 & 0 \end{bmatrix}$$

$-D+0E = -D$ Rule 3C

$$-D = -\begin{bmatrix} -4 & 1 \\ 5 & 0 \\ 2 & -1 \end{bmatrix} = \begin{bmatrix} 4 & -1 \\ -5 & 0 \\ -2 & 1 \end{bmatrix}$$

Section 7.2

Let

$$\mathbf{A} = \begin{bmatrix} 4 & -2 & 3 \\ -2 & 1 & 6 \\ 1 & 2 & 2 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 1 & -3 & 0 \\ -3 & 1 & 0 \\ 0 & 0 & -2 \end{bmatrix}$$

$$\mathbf{C} = \begin{bmatrix} 0 & 1 \\ 3 & 2 \\ -2 & 0 \end{bmatrix}, \quad \mathbf{a} = [1 \quad -2 \quad 0], \quad \mathbf{b} = \begin{bmatrix} 3 \\ 1 \\ -1 \end{bmatrix}.$$

Showing all intermediate results, calculate the following expressions or give reasons why they are undefined:

12. $AA^\top$, A^2, $BB^\top$, B^2

$$AA^T = \begin{bmatrix} 4 & -2 & 3 \\ -2 & 1 & 6 \\ 1 & 2 & 2 \end{bmatrix} \cdot \begin{bmatrix} 4 & -2 & 1 \\ -2 & 1 & 2 \\ 3 & 6 & 2 \end{bmatrix}$$

$$AA^T = \begin{bmatrix} 16+4+9 & -8-2+18 & 4-4+6 \\ -8-2+18 & 4+1+36 & -2+2+12 \\ 4-4+6 & -2+2+12 & 1+4+4 \end{bmatrix}$$

$$AA^T = \begin{bmatrix} 29 & 8 & 6 \\ 8 & 41 & 12 \\ 6 & 12 & 9 \end{bmatrix}$$

$$A^2 = \begin{bmatrix} 4 & -2 & 3 \\ -2 & 1 & 6 \\ 1 & 2 & 2 \end{bmatrix} \cdot \begin{bmatrix} 4 & -2 & 3 \\ -2 & 1 & 6 \\ 1 & 2 & 2 \end{bmatrix}$$

$$A^2 = \begin{bmatrix} 16+4+3 & -8-2+6 & 12-12+6 \\ -8-2+6 & 4+1+12 & -6+6+12 \\ 4-4+2 & -2+2+4 & 3+12+4 \end{bmatrix}$$

$$A^2 = \begin{bmatrix} 23 & -4 & 6 \\ -4 & 17 & 12 \\ 2 & 4 & 19 \end{bmatrix}$$

$$BB^T = B^2 = \begin{bmatrix} 1 & -3 & 0 \\ -3 & 1 & 0 \\ 0 & 0 & -2 \end{bmatrix}^2$$

B is symetric

There fore $B = B^T$

$BB^T = B^2$

$$= \begin{bmatrix} 1 & -3 & 0 \\ -3 & 1 & 0 \\ 0 & 0 & -2 \end{bmatrix} \begin{bmatrix} 1 & -3 & 0 \\ -3 & 1 & 0 \\ 0 & 0 & -2 \end{bmatrix}$$

$$= \begin{bmatrix} 1+9+0 & -3-3+0 & 0+0+0 \\ -3-3+0 & 9+1+0 & 0+0+0 \\ 0+0+0 & 0+0+0 & 0+0+4 \end{bmatrix}$$

$$= \begin{bmatrix} 10 & -6 & 0 \\ -6 & 10 & 0 \\ 0 & 0 & 4 \end{bmatrix}$$

14. $3\mathbf{A} - 2\mathbf{B}$, $(3\mathbf{A} - 2\mathbf{B})^T$, $3\mathbf{A}^T - 2\mathbf{B}^T$, $(3\mathbf{A} - 2\mathbf{B})^T\mathbf{a}^T$

$$3A - 2B = 3\begin{bmatrix} 4 & -2 & 3 \\ -2 & 1 & 6 \\ 1 & 2 & 2 \end{bmatrix} - 2\begin{bmatrix} 1 & -3 & 0 \\ -3 & 1 & 0 \\ 0 & 0 & -2 \end{bmatrix}$$

$$= \begin{bmatrix} 12 & -6 & 9 \\ -6 & 3 & 18 \\ 3 & 6 & 6 \end{bmatrix} - \begin{bmatrix} 2 & -6 & 0 \\ -6 & 2 & 0 \\ 0 & 0 & -4 \end{bmatrix}$$

$$= \begin{bmatrix} 10 & 0 & 9 \\ 0 & 1 & 18 \\ 3 & 6 & 10 \end{bmatrix}$$

$$(3A - 2B)^T = 3A^T - 2B^T \quad \boxed{\text{Rule 10 b}}$$

$$= \begin{bmatrix} 10 & 0 & 9 \\ 0 & 1 & 18 \\ 3 & 6 & 10 \end{bmatrix}^T$$

$$= \begin{bmatrix} 10 & 0 & 3 \\ 0 & 1 & 6 \\ 9 & 18 & 10 \end{bmatrix}$$

$$(3A - 2B)^T a^T = \begin{bmatrix} 10 & 0 & 3 \\ 0 & 1 & 6 \\ 9 & 18 & 10 \end{bmatrix} \cdot \begin{bmatrix} 1 & -2 & 0 \end{bmatrix}^T$$

$$= \begin{bmatrix} 10 & 0 & 3 \\ 0 & 1 & 6 \\ 9 & 18 & 10 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ -2 \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} 10 & +0 & +0 \\ 0 & -2 & +0 \\ 9 & -36 & +0 \end{bmatrix}$$

$$= \begin{bmatrix} 10 \\ -2 \\ -27 \end{bmatrix}$$

17. $\mathbf{ABC}$, $\mathbf{ABa}$, $\mathbf{ABb}$, $\mathbf{Ca}^T$

$$ABC = (AB)C$$

$$= \left(\begin{bmatrix} 4 & -2 & 3 \\ -2 & 1 & 6 \\ 1 & 2 & 2 \end{bmatrix} \begin{bmatrix} 1 & -3 & 0 \\ -3 & 1 & 0 \\ 0 & 0 & -2 \end{bmatrix} \right) \begin{bmatrix} 0 & 1 \\ 3 & 2 \\ -2 & 0 \end{bmatrix}$$

$$= \begin{pmatrix} 4+6+0 & -12-2+0 & 0+0-6 \\ -2-3+0 & 6+1+0 & 0+0-12 \\ 1-6+0 & -3+2+0 & 0+0-4 \end{pmatrix} \begin{bmatrix} 0 & 1 \\ 3 & 2 \\ -2 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 10 & -14 & -6 \\ -5 & 7 & -12 \\ -5 & -1 & -4 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 3 & 2 \\ -2 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 0-42+12 & 10-28+0 \\ 0+21+24 & -5+14+0 \\ 0-3+8 & -5-2+0 \end{bmatrix}$$

$$= \begin{bmatrix} -30 & -18 \\ 45 & 9 \\ 5 & -7 \end{bmatrix}$$

ABa is not possible

AB is 3×3

a is 1×3

incompatible sizes

$$AB\,b = \begin{bmatrix} 10 & -14 & -6 \\ -5 & 7 & -12 \\ -5 & -1 & -4 \end{bmatrix} \cdot \begin{bmatrix} 3 \\ 1 \\ -1 \end{bmatrix}$$

$$= \begin{bmatrix} 30-14+6 \\ -15+7+12 \\ -15-1+4 \end{bmatrix}$$

$$= \begin{bmatrix} 22 \\ 4 \\ -12 \end{bmatrix}$$

Ca^T is no possible

C is 3×2

a^T is 3×1

Incompatible sizes

Section 7.3

Solve the linear system given explicitly or by its augmented matrix. Show details.

3. $x + y - z = 9$
$8y + 6z = -6$
$-2x + 4y - 6z = 40$

$$\left[\begin{array}{ccc|c} 1 & 1 & -1 & 9 \\ 0 & 8 & 6 & -6 \\ -2 & 4 & -6 & 40 \end{array}\right] \xrightarrow{R_3 + 2R_1 \to R_3} \left[\begin{array}{ccc|c} 1 & 1 & -1 & 9 \\ 0 & 8 & 6 & -6 \\ 0 & 6 & -8 & 58 \end{array}\right]$$

$$\xrightarrow{\frac{1}{8}R_2 \to R_2} \left[\begin{array}{ccc|c} 1 & 1 & -1 & 9 \\ 0 & 1 & \frac{6}{8} & -\frac{6}{8} \\ 0 & 6 & -8 & 58 \end{array}\right] \xrightarrow[R_3 - 6R_2 \to R_3]{R_1 - R_2 \to R_1} \left[\begin{array}{ccc|c} 1 & 0 & -\frac{7}{4} & \frac{39}{4} \\ 0 & 1 & \frac{6}{8} & -\frac{6}{8} \\ 0 & 0 & -\frac{25}{2} & \frac{125}{2} \end{array}\right]$$

$$\xrightarrow{-\frac{2}{25}R_3 \to R_3} \left[\begin{array}{ccc|c} 1 & 0 & -\frac{7}{4} & \frac{39}{4} \\ 0 & 1 & \frac{6}{8} & -\frac{6}{8} \\ 0 & 0 & 1 & -5 \end{array}\right] \xrightarrow[R_1 + \frac{7}{4}R_3 \to R_1]{R_2 - \frac{6}{8}R_3 \to R_2} \left[\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 1 & -5 \end{array}\right]$$

Solution $x = 1$; $y = 3$; $z = -5$

9. $-2y - 2z = -8$
$3x + 4y - 5z = 13$

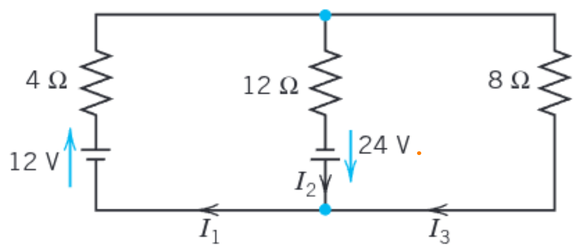
$$\left[\begin{array}{ccc|c} 0 & -2 & -2 & -8 \\ 3 & 4 & -5 & 13 \end{array}\right] \xrightarrow{R_2 \leftrightarrow R_1} \left[\begin{array}{ccc|c} 3 & 4 & -5 & 13 \\ 0 & -2 & -2 & -8 \end{array}\right]$$

$$\xrightarrow{\frac{1}{3}R_1 \to R_1} \left[\begin{array}{ccc|c} 1 & \frac{4}{3} & -\frac{5}{3} & \frac{13}{3} \\ 0 & -2 & -2 & -8 \end{array}\right] \xrightarrow{-\frac{1}{2}R_2 \to R_2} \left[\begin{array}{ccc|c} 1 & \frac{4}{3} & -\frac{5}{3} & \frac{13}{3} \\ 0 & 1 & 1 & 4 \end{array}\right]$$

$$\xrightarrow{R_1 - \frac{4}{3}R_2 \to R_1} \left[\begin{array}{ccc|c} 1 & 0 & -3 & -1 \\ 0 & 1 & 1 & 4 \end{array}\right]$$

Solution $x = 3z - 1$
$y = 4 - z$
$z \in \mathbb{R}$

18.



From the node:

$$I_1 - I_2 - I_3 = 0$$

From the left loop

$$4I_1 + 12I_2 = 36$$

From the right loop

$$12I_2 - 8I_3 = 24$$

Then the augmented matrix

$$\left[\begin{array}{ccc|c} 1 & -1 & -1 & 0 \\ 4 & 12 & 0 & 36 \\ 0 & 12 & -8 & 24 \end{array}\right] \xrightarrow{R_2 - 4R_1 \to R_2} \left[\begin{array}{ccc|c} 1 & -1 & 1 & 0 \\ 0 & 16 & 4 & 36 \\ 0 & 12 & -8 & 24 \end{array}\right]$$

$$\xrightarrow{\frac{1}{16}R_2 \to R_2} \left[\begin{array}{ccc|c} 1 & -1 & 1 & 0 \\ 0 & 1 & \frac{1}{4} & \frac{9}{4} \\ 0 & 12 & -8 & 24 \end{array}\right] \xrightarrow[R_3 - 12R_2 \to R_3]{R_1 + R_2 \to R_1} \left[\begin{array}{ccc|c} 1 & 0 & -\frac{3}{4} & \frac{9}{4} \\ 0 & 1 & \frac{1}{4} & \frac{9}{4} \\ 0 & 0 & -11 & -3 \end{array}\right]$$

$$\xrightarrow{-\frac{1}{11}R_3 \to R_3} \left[\begin{array}{ccc|c} 1 & 0 & -\frac{3}{4} & \frac{9}{4} \\ 0 & 1 & \frac{1}{4} & \frac{9}{4} \\ 0 & 0 & 1 & \frac{3}{11} \end{array}\right] \xrightarrow[R_1 + \frac{3}{4}R_3 \to R_1]{R_2 - \frac{1}{4}R_3 \to R_2} \left[\begin{array}{ccc|c} 1 & 0 & 0 & \frac{27}{11} \\ 0 & 1 & 0 & \frac{24}{11} \\ 0 & 0 & 1 & \frac{3}{11} \end{array}\right]$$

Solution $I_1 = \frac{27}{11}$ $I_2 = \frac{24}{11}$ $I_3 = \frac{3}{11}$

Section 7.4

Find the rank. Find a basis for the row space. Find a basis for the column space. *Hint.* Row-reduce the matrix and its transpose. (You may omit obvious factors from the vectors of these bases.)

2. $\begin{bmatrix} a & b \\ b & a \end{bmatrix}$ Let the matrix $A = \begin{bmatrix} a & b \\ b & a \end{bmatrix}$

Applying reduction:

$$\begin{bmatrix} a & b \\ b & a \end{bmatrix} \xrightarrow{\frac{1}{a}R_1 \to R_1} \begin{bmatrix} 1 & \frac{b}{a} \\ b & a \end{bmatrix} \xrightarrow{R_2 - bR_1 \to R_2} \begin{bmatrix} 1 & \frac{b}{a} \\ 0 & \frac{a^2-b^2}{a} \end{bmatrix}$$

Here I assume $a \neq 0$

Now we have 2 scenarios

→ $a^2 - b^2 \neq 0$ then the matrix can be reduced to Identity Matrix

→ $a^2 - b^2 = 0$ then $a^2 - b^2 = 0$

$(a+b)(a-b) = 0$

$a = b$ or $a = -b$

Here the matrix will lose one independent row

It will be reduced

IF $a = b$ $\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$ If $a = -b$ $\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$

Condensing the info

If $a = b = 0$

Rank is zero : $\text{rank}(A) = 0$

No Row basis : Basis Row $(A) = \emptyset$

No Column basis : Basis Column$(A) = \emptyset$

If $a = b$ and $a \neq 0$

$\text{rank}(A) = 1$

$\text{Basis Row}(A) = \{[1 \ 1]\}$

$\text{Basis Colum}(A) = \left\{\begin{bmatrix} 1 \\ 1 \end{bmatrix}\right\}$

If $a = -b$ and $a \neq 0$

$\text{rank}(A) = 1$

$\text{Basis Row}(A) = \{[1 \ -1]\}$

$\text{Basis Column}(A) = \left\{\begin{bmatrix} 1 \\ -1 \end{bmatrix}\right\}$

If $a \neq b$ and $a, b \neq 0$

$\text{rank}(A) = 2$

$\text{Basis Row}(A) = \{[1 \ 0], [0 \ 1]\}$

$\text{Basis Colum}(A) = \left\{\begin{bmatrix} a \\ b \end{bmatrix}, \begin{bmatrix} b \\ a \end{bmatrix}\right\}$

9. $\begin{bmatrix} 9 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$ Symetrical Matrix A

$$\begin{bmatrix} 9 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix} \xrightarrow[R_4 - R_2 \to R_4]{\frac{1}{9}R_1 \to R_1} \begin{bmatrix} 1 & 0 & \frac{1}{9} & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow{R_3 \leftrightarrow R_2}$$

$$\begin{bmatrix} 1 & 0 & \frac{1}{9} & 0 \\ 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow{R_2 - R_1 \to R_2} \begin{bmatrix} 1 & 0 & \frac{1}{9} & 0 \\ 0 & 1 & \frac{8}{9} & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$\text{Rank}(A) = 3$

$\text{Basis Row}(A) = \{[1 \ 0 \ \frac{1}{9} \ 0], [0 \ 1 \ \frac{8}{9} \ 1], [0 \ 0 \ 1 \ 0]\}$

$\text{Basis Col}(A) = \left\{\begin{bmatrix} 9 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}\right\}$

Show the following:

12. rank $\mathbf{B}^\mathsf{T}\mathbf{A}^\mathsf{T}$ = rank $\mathbf{AB}$. (Note the order!)

Using the theorem 3 (page 284)

rank (A) = rank (A^T)

then

$X = AB$

$X^T = (AB)^T = B^T A^T$

According to the theorem:

rank (X) = rank (X^T)

rank (AB) = rank $(B^T A^T)$

14. If **A** is not square, either the row vectors or the column vectors of **A** are linearly dependent.

Let $A \in \mathbb{R}^{n \times m}$

A is not square $\therefore$ $n \neq m$

→ If $n > m$

There are n rows $\mathbb{R}^m$

At maximum m rows can be independient in $\mathbb{R}^m$
then at least $n-m$ rows are dependent

→ If $m > n$

There are m Columns $\mathbb{R}^n$

At maximum n Columns Can be independient in $\mathbb{R}^n$
then at least $m-n$ Columns are dependient

15. If the row vectors of a square matrix are linearly independent, so are the column vectors, and conversely.

Let $A \in \mathbb{R}^{n \times n}$

The row vectors are linearly independent, I can infer that:

$$\text{Rank}(A) = n$$

Following Theorem 3 (Page 284)

$$\text{Rank}(A) = \text{Rank}(A^T) = n$$

A transpose has the A rows as Columns and the A columns as rows: And as the rank of $A^T = n$ then A must have n linearly independent Columns